IRAS Research Data Management Guidelines

Inka Pieterson Marthe Knijff Zimbo Boudewijns

2026-03-05

Table of contents

Preface

This book serves as a guideline for research data management at the Institute for Risk Assessment Sciences (IRAS) of the Faculty of Veterinary Science. It outlines and defines a set of principles regarding research data and provides practical handlers for researchers in accordance with the regulations established in the Utrecht University (UU) Policy Framework for Research Data Management and the Research Data Management Guidelines developed by the Faculty of Veterinary Medicine. These guidelines can be used as a reference together with existing UU policies, faculty guidelines and/or procedures already established by research groups.

Part I

Introduction

The current chapter describes:

- Chapter 1: how this book is structured.
- Chapter 2: why good research data management is important.
- Chapter 3: where you can find support.

1 Structure of this book

The IRAS Research Data Management guidelines are structured according to the Research Data Life Cycle (Figure 1.1). This cycle can be broadly divided into three phases: before (i.e. planning and designing), during and after research. Research data management (RDM) activities and actions take place during all three phases.

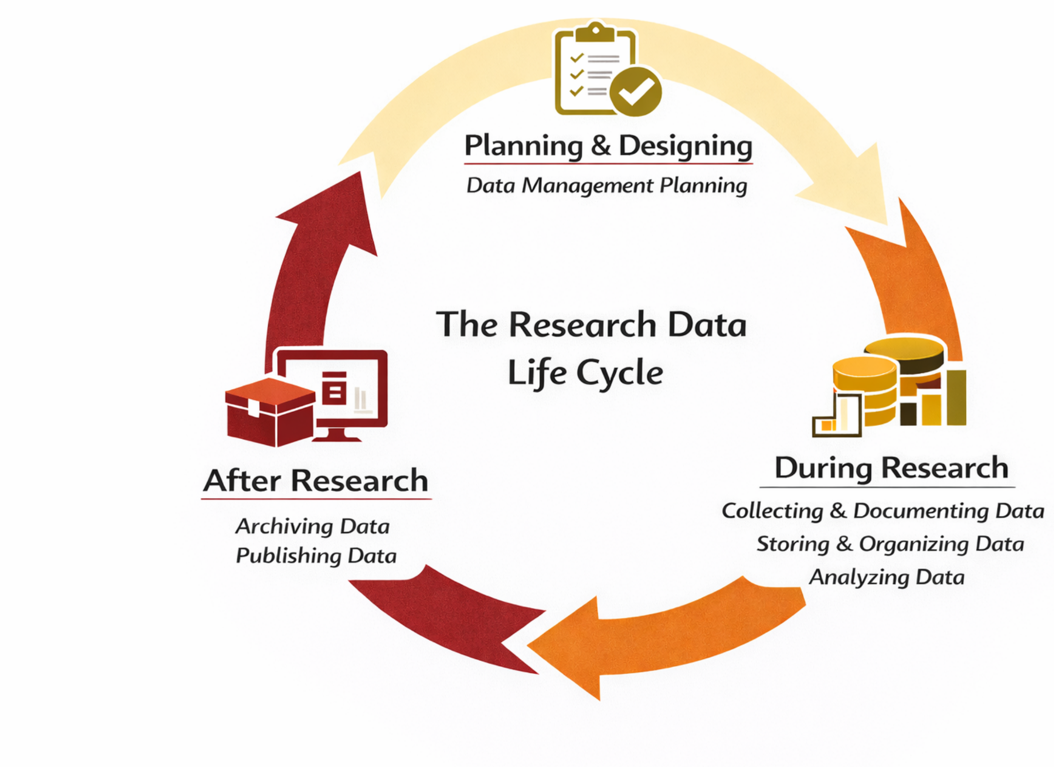

Figure 1.1: The Research Data Life Cycle

2 Importance of good research data management (RDM)

RDM refers to how you handle, organize, and structure your research data throughout the research process. Good RDM is essential for ensuring the integrity and validity of research, facilitating reproducibility and promoting collaboration. It also helps protect sensitive information and comply with regulations. Key reasons why good RDM is important include:

- **Ensuring data quality and security:** proper RDM prevents data loss, protects against unauthorized access, and ensures that research results are reliable and valid.
- **Increasing efficiency:** organizing, documenting, and structuring data saves time and resources by making it easier to find, understand and use data. Not only for others, but also for your future self.
- **Enabling reproducibility and validation:** well-managed, documented data allows other researchers to verify findings, contributing to the overall reliability and validity of scientific knowledge.
- **Allowing for reuse and collaboration:** making data understandable for others facilitates collaboration by enabling researchers to easily share and access data, accelerating scientific discovery.
- **Compliance and integrity:** good RDM ensures compliance with legal (e.g. GDPR when working with personal data), ethical, and funder requirements, and adheres to scientific integrity standards.
- **Improving visibility and impact:** making data FAIR (Findable, Accessible, Interoperable, Reusable) increases citation potential and the overall impact of research.

Proper RDM covers the entire data life cycle, from initial planning and collection to storage, analysis, archiving, and sharing. It is essential for effective Open Science and FAIR practices.

2.1 Open Science

In 2017, in order to accelerate and improve the realization of research results and their societal impact, the UU decided to make the transition to Open Science. Open Science is about increasing the reuse of research, and making sure that publicly funded research is accessible to all. Key to achieving this is adhering to the FAIR principles: ensuring the findings and data behind research results are findable, accessible, interoperable, and reusable (Bruce and Cordewener (2018)).

2.2 FAIR principles

In line with the UU's commitment to Open Science, research data at IRAS is expected to follow the FAIR principles:

- **F - Findable:** data and metadata should be findable, preferably digitally.
- **A - Accessible:** data and metadata should be stored and maintained in such a manner that they are accessible and with clear terms of use.
- **I - Interoperable:** data and metadata data should be stored and maintained in such a manner that the data can be easily opened, exchanged and combined with other datasets.
- **R - Reusable:** data are licensed and described in a manner that facilitates its reuse by other users.

Guidance on how to make your research data FAIR is highlighted in the recommendations provided below. To learn more see the RDM Support guide on how to make your data FAIR.

3 Support

3.1 IRAS Data Management team

The IRAS Data Management team is there to ...

3.2 UU RDM Support team

In addition to the data management team at IRAS, the UU's Research Data Management (RDM) Support team is there to support, advise and assist all researchers at UU with data management, data analysis, research software and code. A multidisciplinary network of data experts, software specialists and research engineers offer their expertise through training, tools, infrastructure, consultancy, and custom-made solutions at every stage of a research project. The support they provide complies with the FAIR principles: findable, accessible, interoperable and reusable. That way, it contributes to Open Science.

> RDM Support workshops and guides
>
> To learn more about how to manage your research data and software, the UU's RDM Support team offers several trainings and workshops, as well as walk-in events. For a complete event agenda, together with information on trainings and workshop, please refer to the RDM Support website.
> Reading guides on diverse RDM topics can also be found on the RDM Support website. The same goes for tools that may be useful.

3.3 Other support services

Besides the IRAS Data Management team and RDM Support team, the following support services exist at UU:

- Research Support Office (RSO) of the Faculty of Veterinary Medicine: each faculty has an RSO. The RSO supports researchers throughout the year in the entire process of acquiring grants. For example: advice, information, reading, examples, budget assistance, data management, ethical questions, administrative data, etc.

- Privacy Officers of the Faculty of Veterinary Medicine: if you are starting a new project or research involving the use of personal data, exchanging data with a third party, or have questions about the secure storage of data or GDPR, please contact the Privacy Officers. You could also contact them for questions about the safe and responsible use of AI.

3.4 Contact

In case you have questions about research data management, please reach out!

> Contact details
>
> IRAS Data Management team: info.irasdm@uu.nl
> RDM Support team: info.rdm@uu.nl
> RSO of the Faculty of Veterinary Medicine: rso-dgk@uu.nl
> Privacy Officers of the Faculty of Veterinary Medicine: privacy-dgk@uu.nl

Part II

Planning and designing

4 Responsibilities

5 Data management plan (DMP)

6 Costs

Part III

During research

7 Collecting and documenting data

7.1 Formdesk

7.2 Qualtrics

7.3 Data formats

7.4 Data documentation

8 Storing and organizing data

8.1 Storing data

8.2 Organizing data

9 Analyzing data

Part IV

After research

10 Archiving data

11 Publishing data

Part V

Legal considerations

12 Legal instruments and agreements, policies and laws

13 Handling personal data

14 Informed consent

15 ...

References

Bruce, R., and B. Cordewener. 2018. "Open Science Is All Very Well but How Do You Make It FAIR in Practice?" *LSE Impact of Social Sciences Blog.* London School of Economics & Political Science. https://blogs.lse.ac.uk/impactofsocialsciences/2018/07/26/open-science-is-all-very-well-but-how-do-you-make-it-fair-in-practice/.

Appendix A

List of mentioned tools

Appendix B

Qualtrics tips & tricks